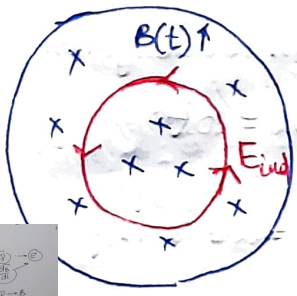
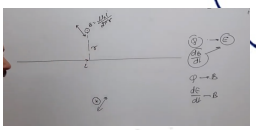
Electromagnetic Waves

→ Non-mechanical waves ⇒ No medium is required.
→ Transverse waves → Disturbance is in plane ⊥ to the velocity of wave.
→ Disturbance are time varying electric and magnetic field
→ Ex = light, X-Ray, γ-rays, radiation, microwave & infrared etc.
→ In vacuum ⇒ $v_{wave} = c = 3\times10^8$ m/s
In any other medium ⇒ $v_{wave} = \frac{c}{n}$ {n = refractive index}

Electric-magnetic Field Coupling:

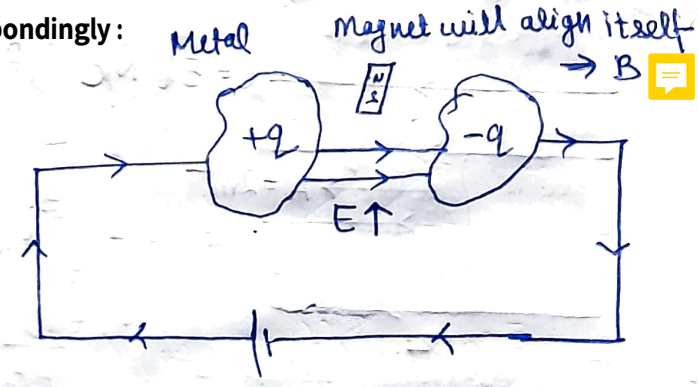
* Time varying magnetic field will induced electric field





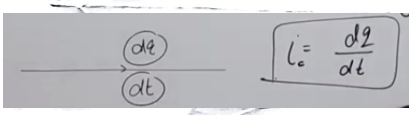
hence, correspondingly :

* Time varying electric field will also create/induce magnetic field.

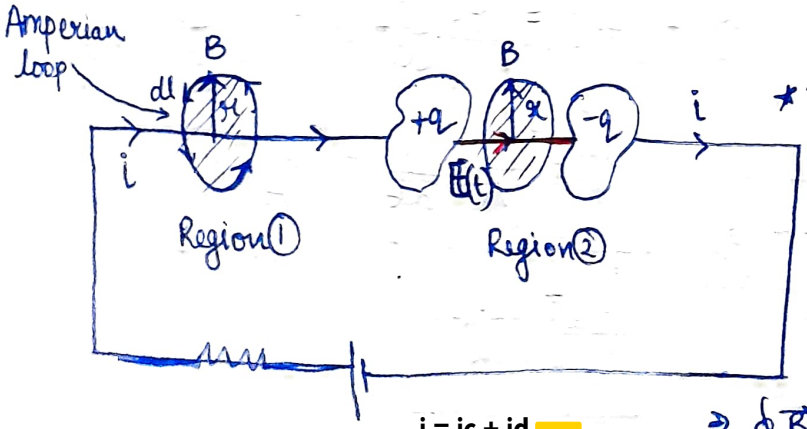


Conduction Current:
→ Current due to motion of charge

$$i_c = \frac{dq}{dt}$$



Maxwell's displacement current (i_d):
→ Not an actual current
→ Not due to motion of charges
→ Pseudo Current
→ It is due to time varying electric field or electric flux.
→ If E is changing with time, ampere's law may not be true. To validate ampere's law, we add an extra term in current i.e. i_d.



Region ① ⇒ $\oint \vec{B}\cdot\vec{dl} = \mu_0 i$

* Experiments showed that B is same for region ① and ②.

But Region ② ⇒ $\oint \vec{B}\cdot\vec{dl} = \mu_0(0) = 0$ wrong.

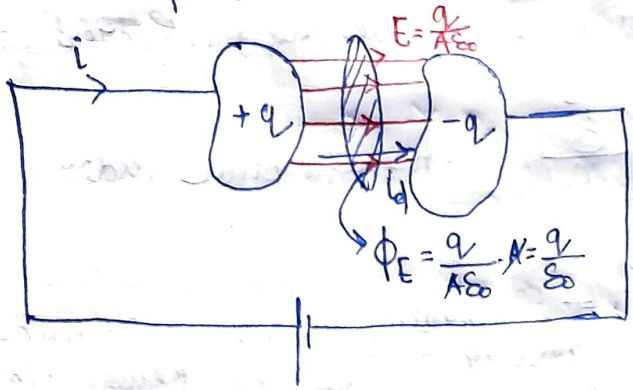
So to validate this law we will do some modification {$i_c = 0$ as no charge flows in ②}

⇒ $\oint \vec{B}\cdot\vec{dl} = \mu_0(i_c + i_d) = \mu_0 i_d$ {As B same ⇒ $i_d = i$}

$i = i_c + i_d$

* We will modify Ampere's law and use i_d only and only if there is time varying electric field.

Formula of i_d:

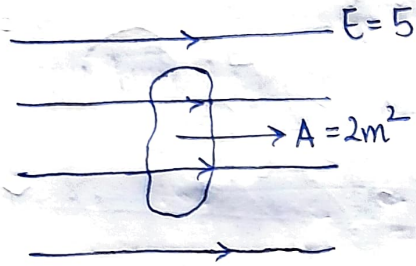


We know,

$$i_d = i = \frac{dq}{dt} = \frac{d}{dt}(\varepsilon_0 \phi_E)$$

$$\therefore * \boxed{i_d = \varepsilon_0 \frac{d(\phi_E)}{dt}}$$

Q

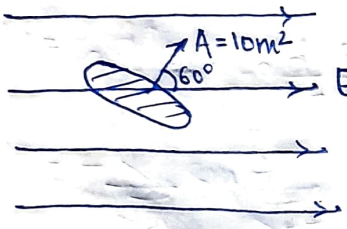


Find i_d through area given

$$\phi_E = 5t \times 2 = 10t$$

$$i_d = \varepsilon_0 \frac{d(10t)}{dt} = 10\varepsilon_0 \text{ Amp}$$

Q

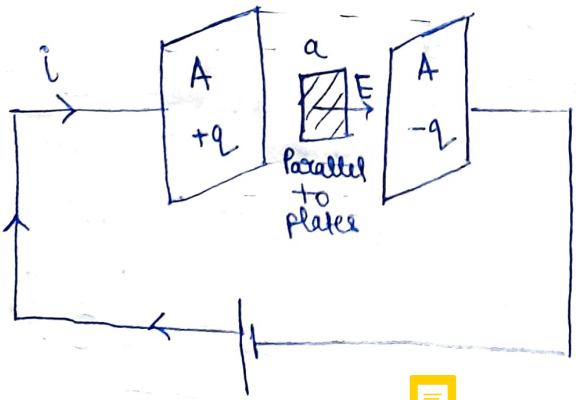


Find i_d through area given

$$\phi_E = 5t^2 \times 10 \times \cos 60° = 25t^2$$

$$i_d = \varepsilon_0 \frac{d(25t^2)}{dt} = 50\varepsilon_0 t \text{ Amp}$$

Q



Find i_d through area a.

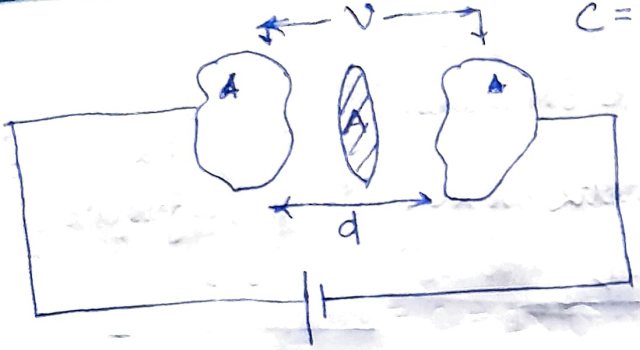
$$\phi_E = \frac{q}{\varepsilon_0 A} a \cos 0°$$

$$i_d = \varepsilon_0 \frac{d}{dt}(\phi_E) = \varepsilon_0 \frac{a}{\varepsilon_0 A} \frac{dq}{dt}$$

$$i_d = \frac{a}{A} \frac{dq}{dt} = \frac{a}{A} i$$

$$\boxed{\frac{i_d}{i} = \frac{a}{A}}$$

Q



$C = 60\mu F$ Find i_d through over section

Given $\frac{dV}{dt} = 5 \frac{Volts}{sec}$

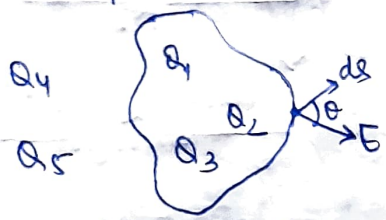
$$i_d = \varepsilon_0 \frac{d\phi_E}{dt} = \varepsilon_0 \frac{d}{dt}\{E \cdot A\}$$

$$= \varepsilon_0 \frac{d}{dt}\left\{\frac{V}{d} A\right\} = \frac{A\varepsilon_0}{d}\frac{dV}{dt}$$

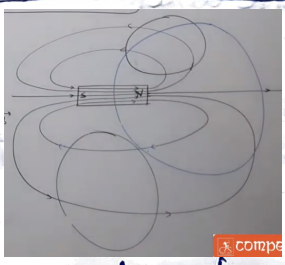
$$i_d = C\frac{dV}{dt} = 60\mu F \times 5 \frac{Volt}{sec} = 300\mu A$$

Maxwell's EM waves equations:

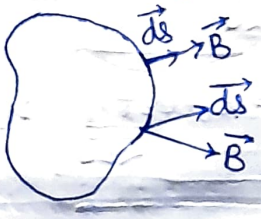
① Gauss law of electrostatics:



$$\phi_E = \oint \vec{E} \cdot d\vec{s} = \frac{Q_{in}}{\varepsilon_0} = \frac{Q_1 + Q_2 + Q_3}{\varepsilon_0}$$



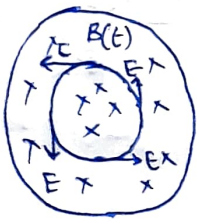
② Gauss law of magnetism:



$$\phi_B = \oint \vec{B} \cdot d\vec{s} = 0$$

Magnetic flux through a closed surface is always zero as no. of lines entering the surface is equal to no. of lines that exit.

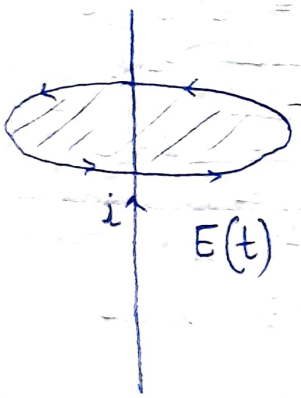
③ Faradays law of EMI:



$$\varepsilon = -\frac{d\phi_B}{dt} = \frac{W}{q} = \frac{\oint q\vec{E} \cdot d\vec{l}}{q} = \oint \vec{E} \cdot d\vec{l}$$

$$\varepsilon = -\frac{d\phi_B}{dt} = \oint \vec{E} \cdot d\vec{l}$$

④ Maxwell Ampere's law: Modified version of ampere's law



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_{in}) = \mu_0 \{i_c + i_d\}$$

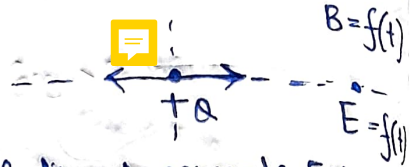
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left\{i_c + \varepsilon_0 \frac{d\phi_E}{dt}\right\}$$

If $i_c = 0$

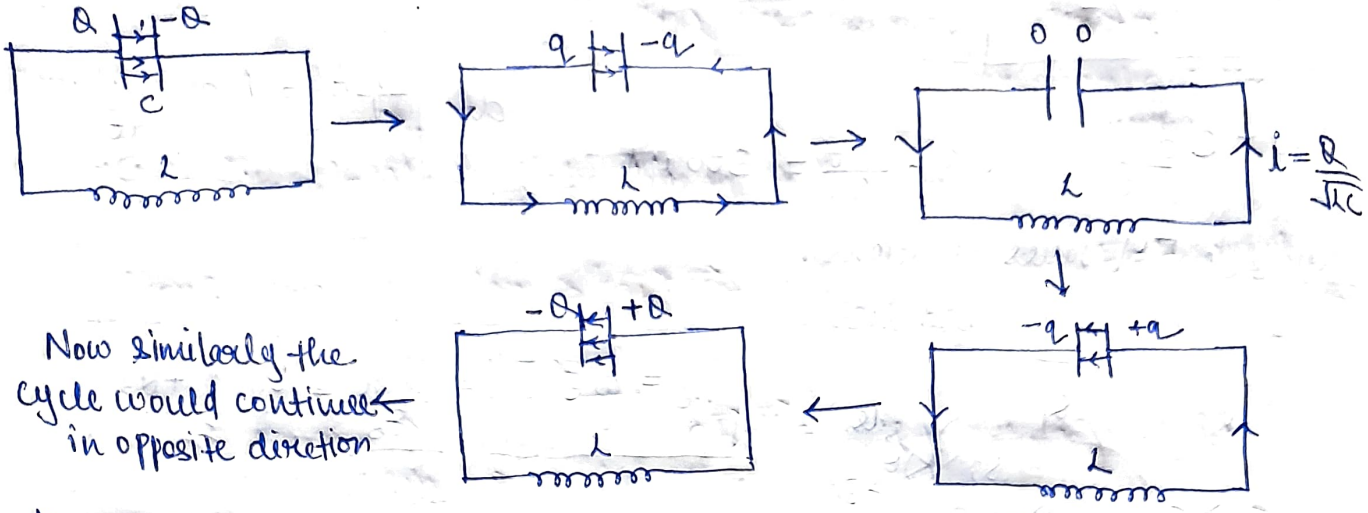
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left\{\varepsilon_0 \frac{d\phi_E}{dt}\right\}$$

Generation of EM waves:

→ Oscillating charged particles generate EM waves.

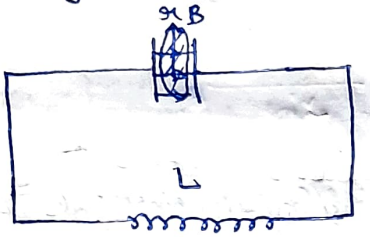


→ An experiment was conducted by hertz on LC oscillation to generate EM wave



Now similarly the cycle would continue in opposite direction

At any time t,



$$q = Q\cos\omega t = Q\cos\frac{t}{\sqrt{LC}}$$

$$E = \frac{q}{A\varepsilon_0} = \frac{Q}{A\varepsilon_0}\cos\frac{t}{\sqrt{LC}} = E_0\cos\frac{t}{\sqrt{LC}} \quad E(t)$$

$$i_d = \varepsilon_0\frac{d\Phi_E}{dt} = \varepsilon_0\frac{d}{dt}\frac{Q}{A\varepsilon_0}\cdot A\cos\frac{t}{\sqrt{LC}}$$

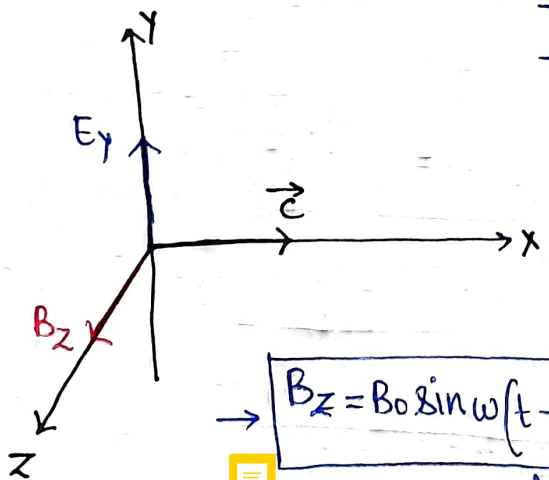
$$i_d = -\frac{Q}{\sqrt{LC}}\sin\frac{t}{\sqrt{LC}}$$

$$B = \frac{\mu_0 i_d}{2\pi x} = \frac{\mu_0 Q}{2\pi x\sqrt{LC}}\sin\left(\frac{t}{\sqrt{LC}}\right) \quad B(t)$$

Equation of EM waves:

→ $\vec{E} \perp \vec{B} \perp \vec{C}$

→ $\vec{E} \times \vec{B}$ provides direction of velocity of wave $\Rightarrow (\vec{E}\times\vec{B}) \parallel \vec{C}$



→ $E_y = E_0 \sin\omega\left(t \mp \frac{x}{c}\right)$ −ve indicates that wave is travelling along +ve x-axis.

→ $B_z = B_0 \sin\omega\left(t - \frac{x}{c}\right)$

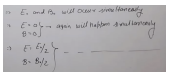
B_0 = Amplitude of magnetic field

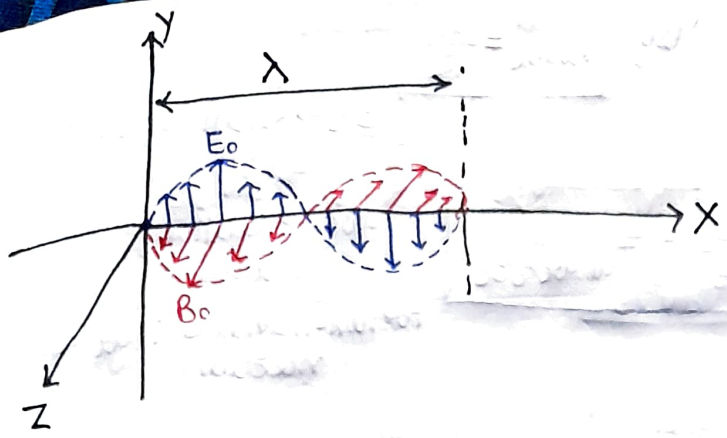
E_0 = Amplitude of electric field

ω = Angular frequency $= 2\pi f = \frac{2\pi}{T}$

C = Speed of E.M. wave.

* $\vec{E}$ and $\vec{B}$ are in same phase.





$$E = E_0 \sin\omega\left(t - \frac{x}{c}\right)$$
$$= E_0 \sin\left(\omega t - \frac{\omega}{c}x\right)$$
$$= E_0 \sin(\omega t - kx) \quad \{k = \frac{\omega}{c}\}$$

$$\boxed{\vec{E} = E_0 \sin\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right)\hat{j}}$$

$$\boxed{\vec{B} = B_0 \sin\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right)\hat{k}}$$

Relation between $\vec{E}$ and $\vec{B}$ in EM wave:

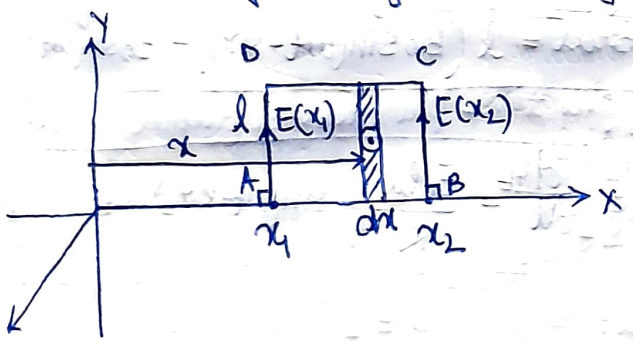
* $\boxed{E_0 = CB_0}$

Multiplying both sides by $\sin(\omega t - kx)$

$$E_0 \sin(\omega t - kx) = cB_0 \sin(\omega t - kx)$$

$$\Rightarrow * \boxed{E = CB} - * \boxed{\vec{E} = \vec{B} \times \vec{C}}$$

Derivation: Using Faraday's law $\Rightarrow \oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$ — (A)



$$E = E_0 \sin(\omega t - kx)$$
$$B = B_0 \sin(\omega t - kx)$$

$$\oint \vec{E} \cdot d\vec{l} = \int_A^B E\,dl\cos 90^\circ + \int_B^C E(x_2)\,dl + \int_C^D E\,dl\cos 90$$
$$+ \int -E(x_1)\,dl$$
$$= 0 + E(x_2)l + 0 - E(x_1)l$$

$$\oint \vec{E} \cdot d\vec{l} = [E(x_2) - E(x_1)]l$$

$$\oint \vec{E} \cdot d\vec{l} = l\{E_0 \sin(\omega t - kx_2) - E_0 \sin(\omega t - kx_1)\} \quad \text{— (1)}$$

$$\phi_B = \int_{x_1}^{x_2} B_0 \sin(\omega t - kx)\,l\,dx$$

$$\phi_B = -\frac{B_0 l}{k}\left(\cos(\omega t - kx)\right)_{x_1}^{x_2}$$

$$\phi_B = \frac{B_0 l}{k}\{\cos(\omega t - kx_2) - \cos(\omega t - kx_1)\} \quad \text{— (2)}$$

Put (1) and $\frac{d}{dt}$(2) in (A)

$$lE_0\{\sin(\omega t - kx_2) - \sin(\omega t - kx)\} = \frac{B_0 l\omega}{k}\{\sin(\omega t - kx_2) - \sin(\omega t - kx_1)\}$$

$$= E_0 = \frac{B_0\omega}{k} = \frac{B_0\omega}{\omega/c} = \frac{B_0\omega c}{\omega} = B_0 c \quad \therefore \quad \boxed{E_0 = CB_0}$$

Speed of EM wave:

① In vacuum, $\boxed{c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}}$

Also, $\boxed{v_{wave} = \frac{\omega}{k}}$

② In any other medium, $\boxed{v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0 \mu_r \varepsilon_0 \varepsilon_r}}}$

$\varepsilon_r \rightarrow$ Relative permittivity, $\varepsilon_r \geq 1$

$\mu_r \rightarrow$ Relative permeability, $\mu_r > 0$

For non-magnetic medium: $\mu_r \approx 1$

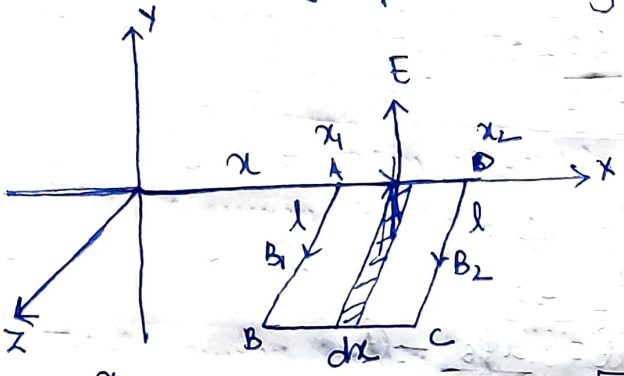
Divide ① and ②

$$\boxed{\frac{c}{v} = \sqrt{\mu_r \varepsilon_r} = n}$$ n = Refractive index of medium

Derivation: Using ampere's law $\Rightarrow \oint \vec{B}\cdot d\vec{l} = \mu_0 \left\{ \varepsilon_0 \frac{d\phi_E}{dt} \right\}$ — (A)

$E = E_0 \sin(\omega t - kx)$

$B = B_0 \sin(\omega t - kx)$



$$\oint \vec{B}\cdot d\vec{l} = \int_A^B B_1 dl + \int_B^C B dl \cos 90^\circ + \int_C^D -B_2 dl + \int_D^A B dl \cos 90^\circ$$

$$= B_1 l + 0 - B_2 l + 0$$

$$= (B_1 - B_2) l$$

$$\phi_E = \int_{x_1}^{x_2} (E_0 \sin(\omega t - kx)) l \, dx$$

$$\boxed{\oint \vec{B}\cdot d\vec{l} = l\left(B_0 \sin(\omega t - kx_1) - B_0 \sin(\omega t - kx_2)\right)} \rightarrow ①$$

$$\boxed{\phi_E = \frac{E_0 l}{k}\left(\cos(\omega t - kx_2) - \cos(\omega t - kx_1)\right)}$$

$$\boxed{\frac{d\phi_E}{dt} = \frac{E_0 l \omega}{k}\left(-\sin(\omega t - kx_2) + \sin(\omega t - kx_1)\right)} \rightarrow ②$$

Put ① and ② in (A)

$$B_0 l \left(\sin(\omega t - kx_1) - \sin(\omega t - kx_2)\right) = \frac{\mu_0 \varepsilon_0 E_0 l \omega}{k}\left(\sin(\omega t - kx_1) - \sin(\omega t - kx_2)\right) \quad ①$$

$$B_0 = \frac{\mu_0 \varepsilon_0 E_0 \omega}{k} \Rightarrow \frac{1}{\mu_0 \varepsilon_0} = c\frac{E_0}{B_0} = c^2 \quad ②$$

$$\therefore \boxed{c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}}$$

Q An EM wave travels along +ve x-axis with speed c. Given $\lambda = 5\,mm$, $E_y = 30\ \frac{V}{m}$ (amplitude)

Find ① $B_0 = \frac{E_0}{c} = \frac{30}{3\times10^8} = 10^{-7}$ Tesla

② $\vec{E_y} = E_0 \sin(\omega t - kx) = 30 \sin(12\times10^{10}\pi t - 400\pi x + \theta)\hat{j}$

③ $\vec{B_z} = 10^{-7} \sin(12\times10^{10}\pi t - 400\pi x + \theta)\hat{k}$

$k = \frac{2\pi}{\lambda} = \frac{2\pi}{5\times10^{-3}} = \frac{2000\pi}{5} = 400\pi$

$k = \frac{\omega}{c} \Rightarrow \omega = kc = 400\pi \times 3\times10^8$

Q Given equation of EM wave, $\vec{E_y} = 300 \sin(6\times10^8 t - 4x)$.

Find ① $B_0 = \frac{E_0}{c} = \frac{300}{3\times10^8} = 10^{-6}\,T$ ② $\omega = 6\times10^8 \frac{rad}{sec}$ ③ $k = 4\,rad/m$

④ $V_{wave} = \frac{\omega}{k} = \frac{6\times10^8}{4} = \frac{3}{2}\times10^8\,m/s$ ⑤ Refractive index of medium $= n = \frac{c}{v} = \frac{3\times10^8}{\frac{3}{2}\times10^8} = 2$

⑥ If medium is non-magnetic, find its dielectric constant.

$n = \sqrt{\varepsilon_r \mu_r}$ For non-magnetic medium $\mu_r = 1$

$2 = \sqrt{\varepsilon_r \times 1} \Rightarrow \boxed{\varepsilon_r = 4}$

Q EM wave $\Rightarrow \vec{E_y} = 600\sin(\omega t - kx)\hat{j}$ electron: $\vec{v} = 2\times10^7\,m/s\,\hat{j}$

Find ① Maximum electric force on electron.

$F_{max} = eE_0 = 1.6\times10^{-19}\times600\,N$

② Maximum magnetic force on e.

$B_0 = \frac{E_0}{c} = \frac{600}{3\times10^8} = 200\times10^{-8} = 2\times10^{-6}$

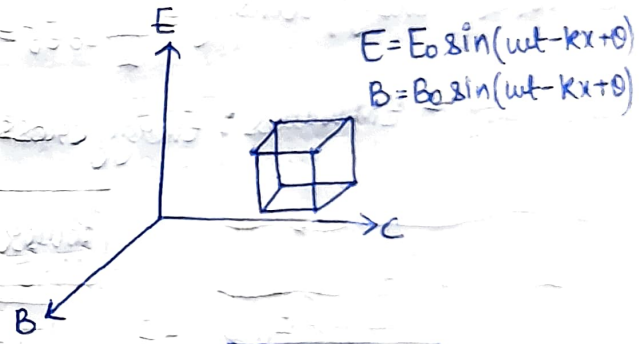
$\frac{f_m}{f_e} = \frac{evB_0}{eE_0} = \frac{v}{c}$

$\vec{F_m} = q(\vec{v}\times\vec{B}) = e\left(2\times10^7\hat{j} \times 2\times10^{-6}\hat{k}\right) = \left(1.6\times10^{-19}\times2\times10^7\times2\times10^{-6}\right)\hat{i}$

$F_m = 6.4\times10^{-18}$

Energy density of EM waves:

- Energy / Volume
- Unit = Joule/m^3
- Dimension = $[ML^{-1}T^{-2}]$



$E = E_0\sin(\omega t - kx + \theta)$

$B = B_0\sin(\omega t - kx + \theta)$

① $\left(\frac{dU}{dV}\right)_{elec} = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 E_0^2 \sin^2(\omega t - kx + \theta)$

$\left(\frac{dU}{dV}\right)_{elec_{Avg}} = \frac{1}{2}\varepsilon_0 E_0^2 \times \frac{1}{2} = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{2}\varepsilon_0\left(\frac{E_0}{\sqrt{2}}\right)^2 = \frac{1}{2}\varepsilon_0 (E_{rms})^2$

$\boxed{E_{rms} = \frac{E_0}{\sqrt{2}}}$ $\boxed{B_{rms} = \frac{B_0}{\sqrt{2}}}$

② $\left(\frac{dU}{dV}\right)_B = \frac{B^2}{2\mu_0} = \frac{B_0^2\sin^2(\omega t - kx + \theta)}{2\mu_0}$

$\left(\frac{dU}{dV}\right)_{B_{avg}} = \frac{B_0^2}{2\mu_0}\times\frac{1}{2} = \frac{B_0^2}{4\mu_0} = \frac{\left(\frac{B_0}{\sqrt{2}}\right)^2}{2\mu_0} = \frac{B_{rms}^2}{2\mu_0}$

$\therefore$

$$\left(\frac{dU}{dV}\right)_E = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{2}\varepsilon_0 E_{rms}^2$$

$$\left(\frac{dU}{dV}\right)_B = \frac{B_0^2}{4\mu_0} = \frac{B_{rms}^2}{2\mu_0}$$

Ex. Prove that in EM wave, magnetic energy density and electric energy density are equal.

$$\left(\frac{dU}{dV}\right)_E = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{4}\varepsilon_0 B_0^2 c^2 = \frac{B_0^2 \varepsilon_0}{4} \times \frac{1}{\mu_0 \varepsilon_0} = \frac{B_0^2}{4\mu_0} = \left(\frac{dU}{dV}\right)_B$$

$$\therefore \boxed{\left(\frac{dU}{dV}\right)_E = \left(\frac{dU}{dV}\right)_B}$$

Now,

$$\text{Total energy density} = \frac{dU}{dV} = \left(\frac{dU_e}{dV} + \frac{dU_B}{dV}\right)$$

$$= \frac{1}{4}\varepsilon_0 E_0^2 + \frac{B_0^2}{4\mu_0} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}$$

$$\therefore \boxed{\text{Total energy density} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}}$$

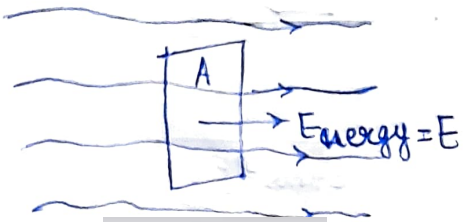
Q EM wave: $E = 200 \sin\omega\left(t - \frac{x}{c}\right)\hat{j}$. Find

① Electric energy density $= \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{4} \times 8.85 \times 10^{-12} \times (200)^2 = 8.85 \times 10^{-8}\ J/m^3$

② Magnetic energy density $= \frac{B_0^2}{4\mu_0} = 8.85 \times 10^{-8}\ J/m^3$

③ Total energy density $= \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0} = 17.7 \times 10^{-8}\ J/m^3$

Intensity of EM waves: Energy crossing p.u time p.u cross-sectional area.



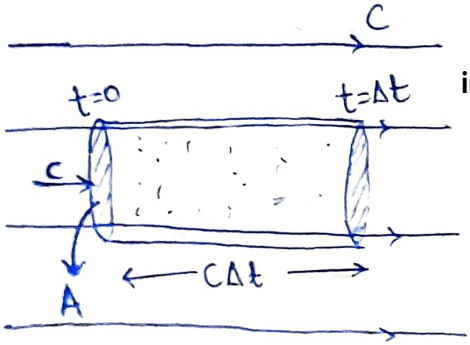
E - energy cross
Δt - time

$$\boxed{\text{Power} = \frac{E}{\Delta t}}$$

$$\boxed{\text{Intensity} = \frac{\text{Power}}{A} = \frac{E}{\Delta t A}}$$

→ Unit = $J/m^2\text{-sec}$ = $Watt/m^2$

→ Dimension = $[MT^{-3}]$



$$\text{Energy Stored} = \left(\frac{dU}{dV}\right) \times \text{Volume}$$

in this cylindrical volume

$$E = \frac{1}{2}\varepsilon_0 E_0^2 \times AC\Delta t$$

$$\text{Power} = \frac{E}{\Delta t} = \frac{1}{2}\varepsilon_0 E_0^2 AC$$

$$\text{Intensity} = \frac{\text{Power}}{A} = \frac{1}{2}\varepsilon_0 E_0^2 C$$

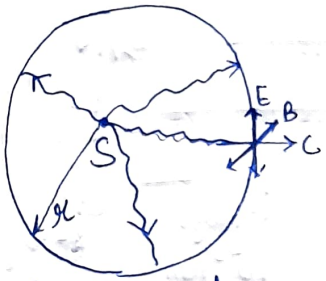
$$\therefore \boxed{I = \frac{1}{2}\varepsilon_0 E_0^2 C = \frac{B_0^2}{2\mu_0} C}$$

Q EM wave: $E = 200 \sin(6\times10^8 t - 2x)$. Find intensity of wave.

$$I = \frac{1}{2}\varepsilon_0 E_0^2 C = \frac{1}{2}\times 8.85\times10^{-12}\times 40000\times\frac{6\times10^8}{2} = 8.85\times 6 = 53.10 \text{ watt/m}^2$$

Ex EM wave $B = (6\times10^{-6})\sin(6\times10^8 t - 2x)$
Find intensity of this wave.
$I = \frac{B_0^2}{2\mu_0} C$
$= \frac{(6\times10^{-6})^2}{2\times 4\pi\times10^{-7}} \times 3\times10^8 \frac{W}{m^2}$

Point Source of EM wave:
Let S be point source of power P



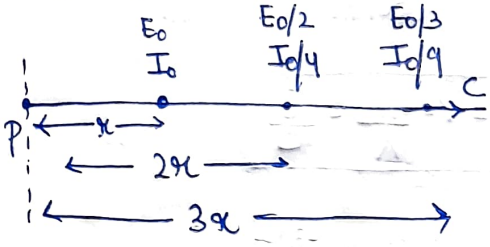
Imaginary sphere

$$I = \frac{P}{4\pi r^2} = \frac{1}{2}\varepsilon_0 E_0^2 C = \frac{B_0^2}{2\mu_0} C$$

$$E_0 = \frac{1}{r}\sqrt{\frac{P_0}{2\pi\varepsilon_0 C}} \qquad B_0 = \frac{1}{r}\sqrt{\frac{\mu_0 P}{2\pi C}}$$

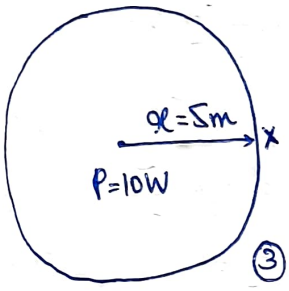
$$E_0 \propto \frac{1}{r} \qquad B_0 \propto \frac{1}{r} \qquad I \propto \frac{1}{r^2}$$

Ex



At $d = x$, $I = I_0$, $E = E_0$, $B = B_0$
At $d = 2x$, $I = I_0/4$, $E = E_0/2$, $B = B_0/2$
At $d = 3x$, $I = I_0/9$, $E = E_0/3$, $B = B_0/3$

Q



Find ① I at X ② E_0 at X ③ B_0 at X

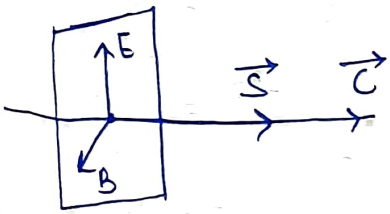
① $I = \frac{10W}{4\pi(5m)^2} = \frac{10W}{100\pi m^2} = \frac{0.1}{\pi}\frac{W}{m^2}$

② $I = \frac{1}{2}\varepsilon_0 E_0^2 C \Rightarrow E_0^2 = \sqrt{\frac{2I}{\varepsilon_0 C}} = \sqrt{\frac{2\times0.1}{8.85\times10^{-12}\times3\times10^8\times\pi}}$

$E_0 = \sqrt{24}$ N/C

③ $B_0 = \frac{E_0}{C} = \frac{\sqrt{24}}{3\times10^8}$ T

Poynting Vector ($\vec{S}$):
↳ Magnitude = Intensity
↳ Direction → towards propogation of wave ($\hat{C}$)



$$\vec{S} = I\hat{C} = \frac{EB}{\mu_0}\hat{C}$$

$$\vec{S} = \frac{1}{\mu_0}(\vec{E}\times\vec{B})$$

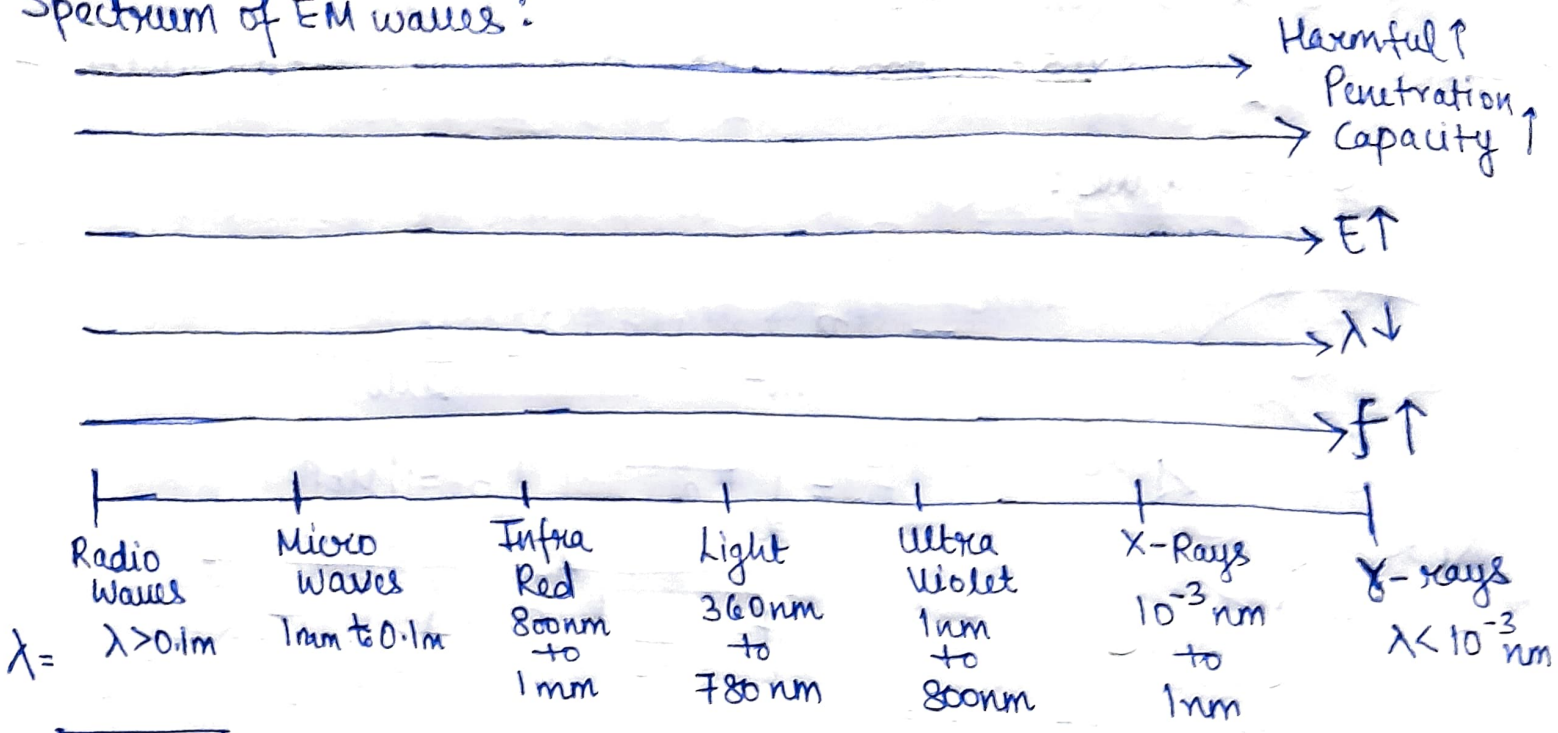
$$I = \frac{1}{2}\varepsilon_0 E_0^2 C = \varepsilon_0\left(\frac{E_0}{\sqrt{2}}\right)^2 C = \varepsilon_0 E_{rms}^2 C$$

$$I = \varepsilon_0 E_{rms}^2 C = \varepsilon_0 E^2 C = (\varepsilon_0 C)E(BC)$$

$$= \varepsilon_0 EBC^2 = \frac{\varepsilon_0 EB}{\mu_0\varepsilon_0} = \frac{EB}{\mu_0}$$

{E and B are rms values}

Spectrum of EM waves:



$$\lambda = \frac{c}{f} \Rightarrow \lambda \propto \frac{1}{f}$$

$$\text{Momentum} = P = \frac{h}{\lambda} = \frac{E}{c}$$

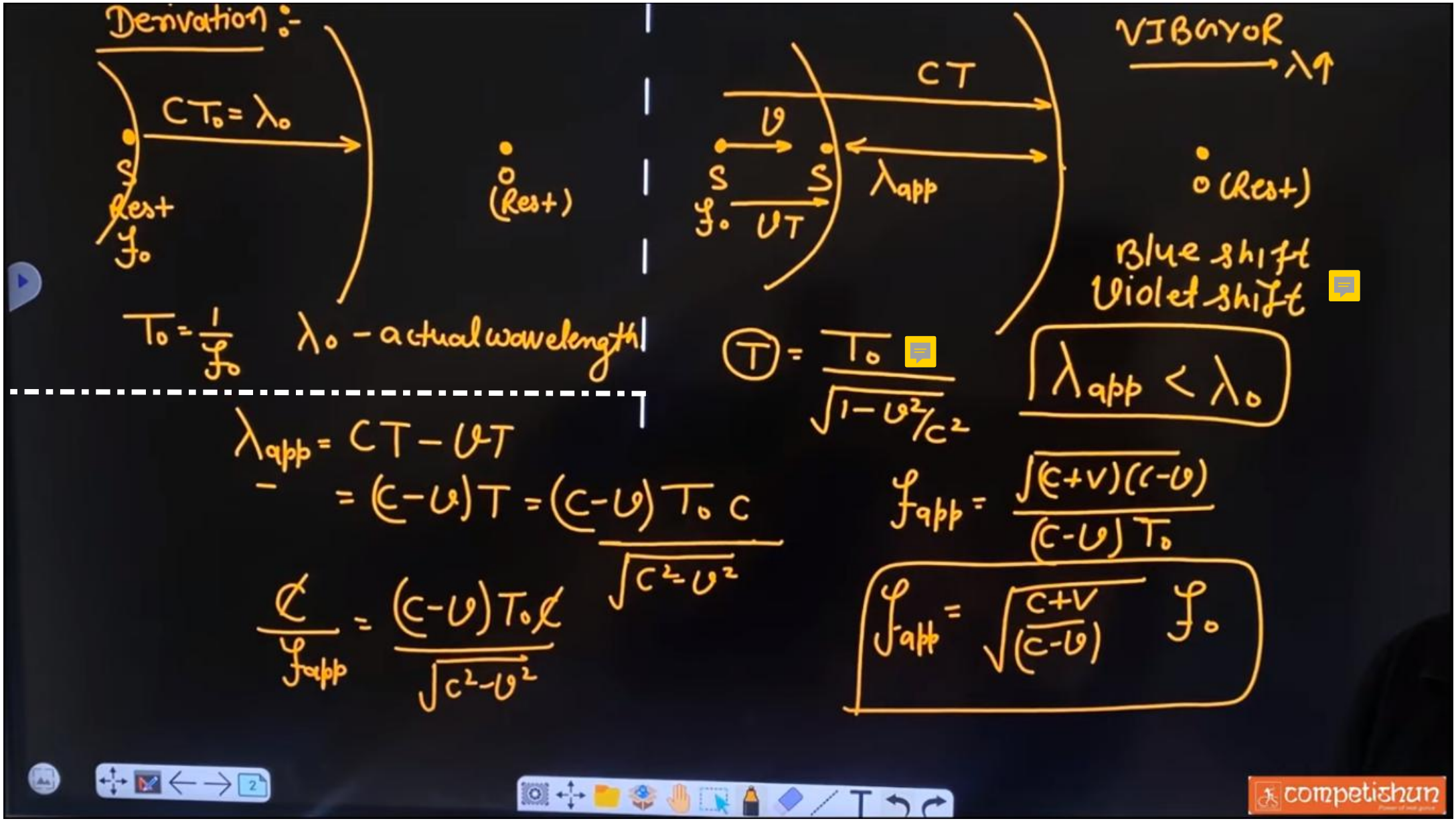
$$E = hf = \frac{hc}{\lambda}$$ {Energy of 1 Photon} $\{h = 6.626 \times 10^{-34}\ J\text{-}s\}$

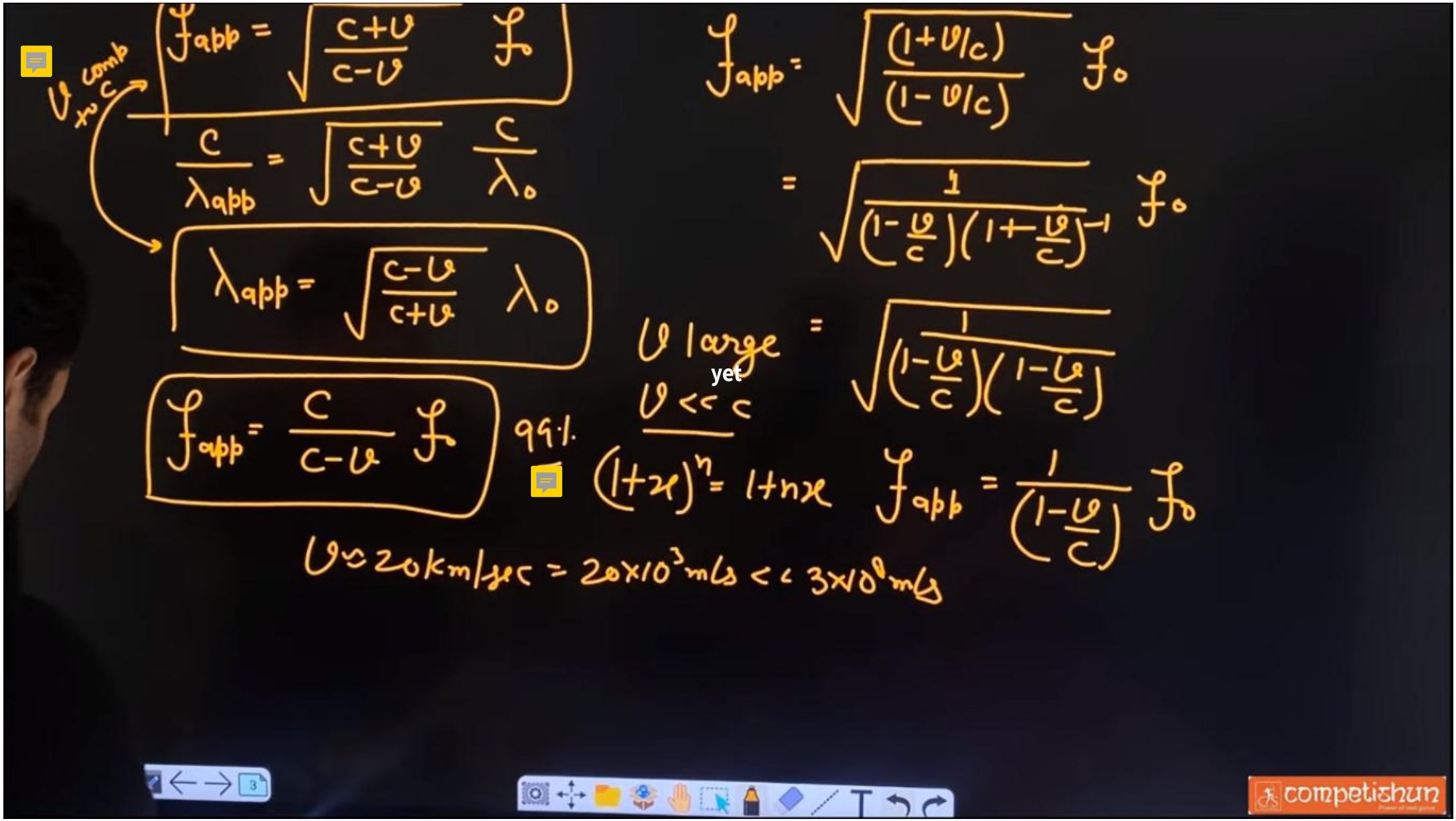
Relativistic Doppler effect {Doppler effect in light}

↳ It is change in frequency and wavelength of light, caused by the relative motion b/w source and observer, when taking into account effects described by "special theory of relativity"

Non Relativistic Doppler effect (Sound)	Relativistic doppler effect (light/EMW)
① Medium is Reference point	Source or observer is reference
② 4 equations	only 2 eqns
③ Classical mechanics (Kinematics)	Special theory of Relativity (Time dilation)
④ $f_{app} = \left(\frac{c \pm v_o}{c \pm v_s}\right) f_o$	$f_{app} = \sqrt{\frac{c+v}{c-v}}\, f_o$ (v ←) ; $f_{app} = \sqrt{\frac{c-v}{c+v}}\, f_o$ (→ v)
⑤ v_o v_s	⑤ v = Relative velocity along the line joining them

competishun


Derivation :-
$CT_0 = \lambda_0$
S
Rest
f_0
(Rest)
$T_0 = \frac{1}{f_0}$
λ_0 - actual wavelength
$\lambda_{app} = CT - vT$
$= (C-v)T = \frac{(C-v)T_0 C}{\sqrt{C^2-v^2}}$
$\frac{C}{f_{app}} = \frac{(C-v)T_0 C}{\sqrt{C^2-v^2}}$
CT
v
S
S
λ_{app}
f_0
vT
$T = \frac{T_0}{\sqrt{1-v^2/C^2}}$
VIBGYOR
$\lambda \uparrow$
(Rest)
Blue shift
Violet shift
$\lambda_{app} < \lambda_0$
$f_{app} = \frac{\sqrt{(C+V)(C-v)}}{(C-v)T_0}$
$f_{app} = \sqrt{\frac{C+V}{(C-v)}} f_0$


$f_{app} = \sqrt{\frac{c+v}{c-v}}\, f_0$
v comb $+ve$ c
$\frac{c}{\lambda_{app}} = \sqrt{\frac{c+v}{c-v}}\, \frac{c}{\lambda_0}$
$\lambda_{app} = \sqrt{\frac{c-v}{c+v}}\, \lambda_0$
$f_{app} = \frac{c}{c-v} f_0$
991.
$f_{app} = \sqrt{\frac{(1+v/c)}{(1-v/c)}}\, f_0$
$= \sqrt{\frac{1}{(1-\frac{v}{c})(1+\frac{v}{c})^{-1}}}\, f_0$
v large
yet
$v << c$
$= \sqrt{\frac{1}{(1-\frac{v}{c})(1-\frac{v}{c})}}$
$(1+x)^n = 1+nx$
$f_{app} = \frac{1}{(1-\frac{v}{c})} f_0$
$v \simeq 20\,km/sec = 20\times10^3\,m/s << 3\times10^8\,m/s$
competishun

$$f_{app} = \frac{c}{c-v} f_0$$

$$\frac{c}{\lambda_{app}} = \frac{c}{c-v} \frac{c}{\lambda_0}$$

$$\boxed{\lambda_{app} = \left(\frac{c-v}{c}\right)\lambda_0}$$

$$\boxed{\Delta\lambda = \frac{v\lambda_0}{c} \downarrow}$$ Rem!

$$\boxed{\frac{\Delta\lambda}{\lambda_0} = \frac{v}{c}}$$

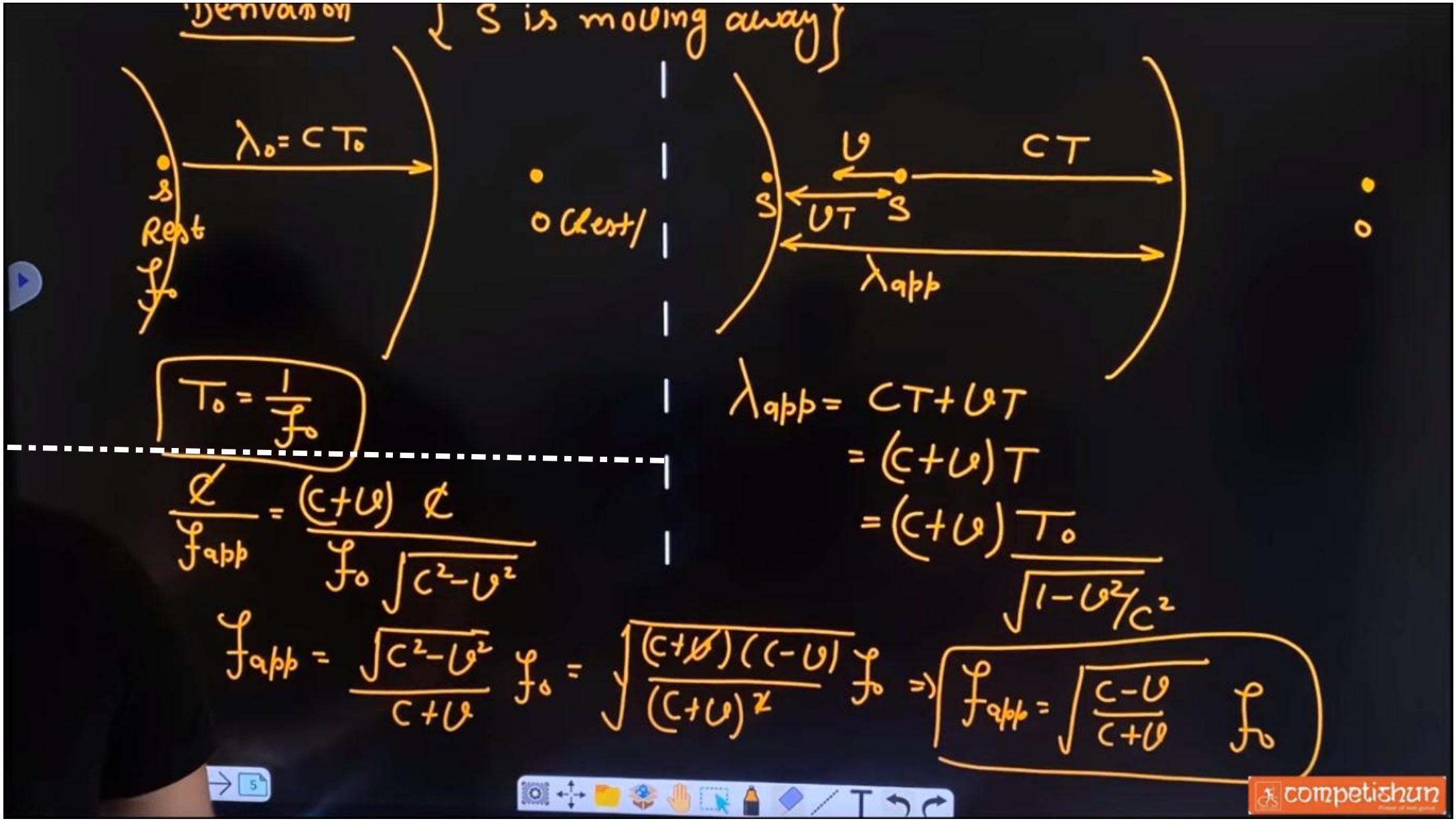
$\lambda_{app} < \lambda_0$ Violet / Blue shift

$$\lambda_{app} - \lambda_0 = \frac{c-v}{c}\lambda_0 - \lambda_0$$

$$= \lambda_0\left(\frac{c-v-c}{c}\right) = \ominus\frac{v\lambda_0}{c}$$

decrease

competishun


Derivation
{ S is moving away }
$\lambda_0 = cT_0$
S
Rest
f
O (Rest)
v
cT
S
vT
S
λ_{app}
$T_0 = \frac{1}{f_0}$
$\lambda_{app} = cT + vT$
$= (c+v)T$
$= (c+v)\frac{T_0}{\sqrt{1-v^2/c^2}}$
$\frac{c}{f_{app}} = \frac{(c+v)\,c}{f_0\sqrt{c^2-v^2}}$
$f_{app} = \frac{\sqrt{c^2-v^2}}{c+v} f_0 = \sqrt{\frac{(c+v)(c-v)}{(c+v)^2}} f_0 \Rightarrow f_{app} = \sqrt{\frac{c-v}{c+v}}\, f_0$
competishun

$$f_{app} = \sqrt{\frac{c-v}{c+v}}\, f_0$$

$$\frac{c}{\lambda_{app}} = \sqrt{\frac{c-v}{c+v}}\,\frac{c}{\lambda_0}$$

$$\boxed{\lambda_{app} = \sqrt{\frac{c+v}{c-v}}\,\lambda_0}$$

$$f_{app} = \sqrt{\frac{(1-v/c)}{(1+v/c)}}\, f$$

$$= \sqrt{\frac{1}{(1+\frac{v}{c})(1-\frac{v}{c})^{-1}}}\, f_0$$

$\lambda_{app} > \lambda_0$

Red shift

$$\boxed{f_{app} = \frac{c}{c+v} f_0}$$

$$\underline{\lambda_{app} = \left(\frac{c+v}{c}\right)\lambda_0}$$

T.P.

if $v << c$

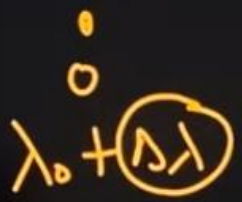
$$f_{app} = \sqrt{\frac{1}{(1+\frac{v}{c})^2}}\, f_0$$

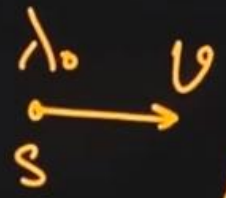
$$f_{app} = \frac{1}{(1+\frac{v}{c})} f_0$$

(99%)

competishun

Red shift

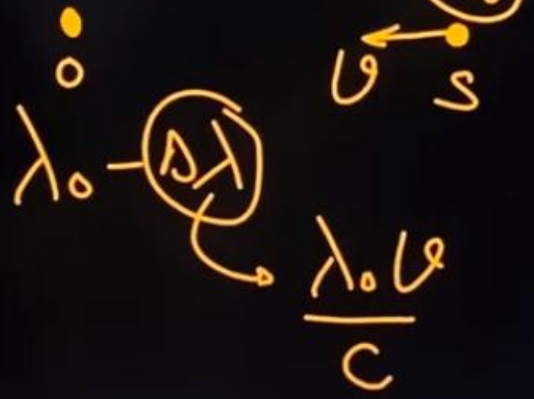




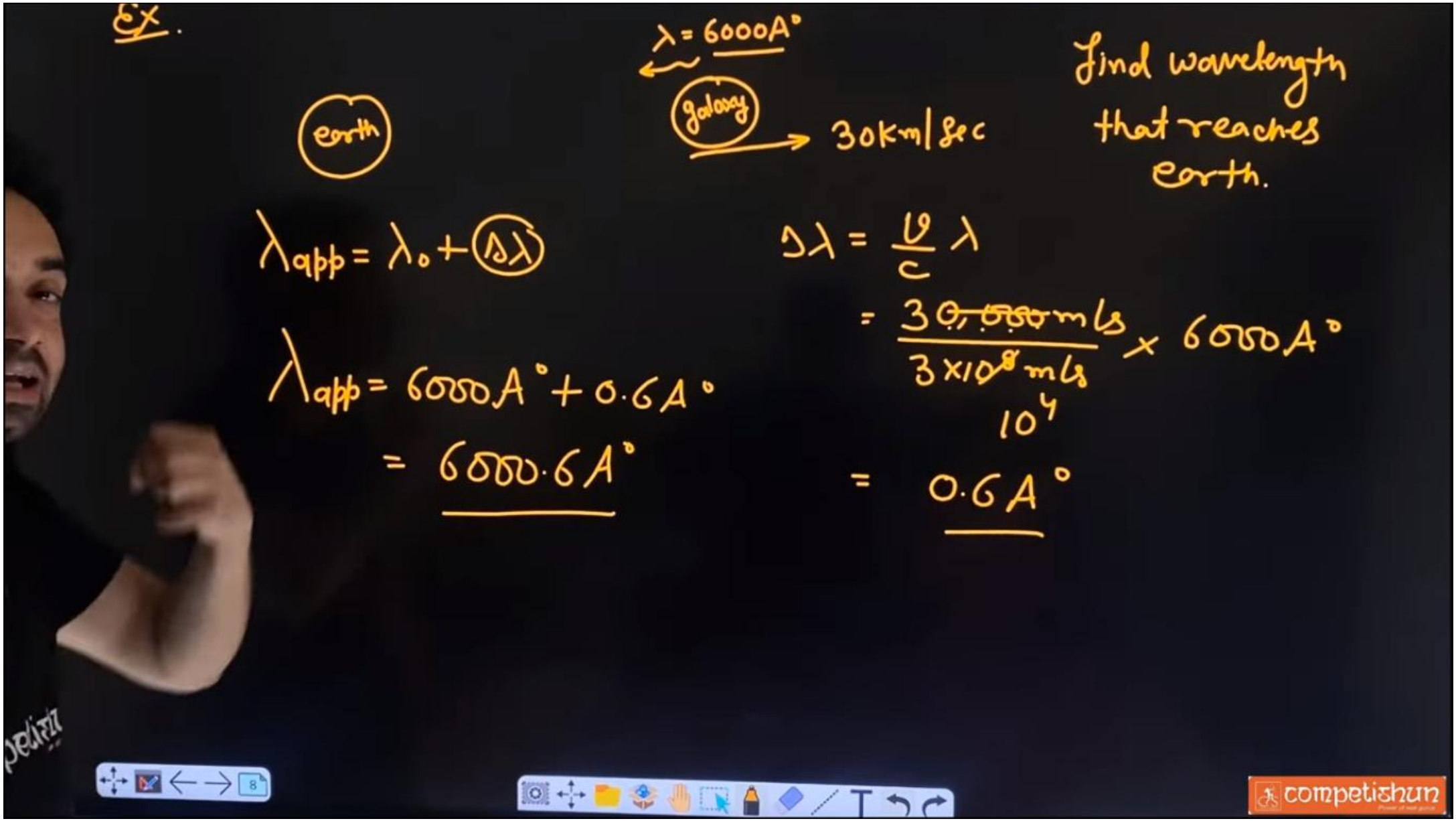
$$\Delta\lambda = \frac{c+v}{c}\lambda_0 - \lambda_0$$

$$\Delta\lambda = \frac{v}{c}\lambda_0$$

Blue shift

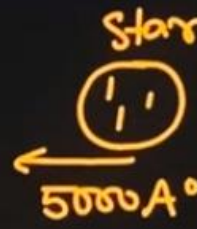


competishun


Ex.
λ = 6000A°
galaxy
30km/sec
Find wavelength that reaches earth.
earth
λapp = λ0 + Δλ
Δλ = v/c λ
= 30,000 m/s / 3×10^8 m/s × 6000A°
10^4
= 0.6A°
λapp = 6000A° + 0.6A°
= 6000.6A°

Ex





wavelength that reaches earth is 4997.5A°. Find speed of star.

$$\Delta\lambda = 5000 - 4997.5$$

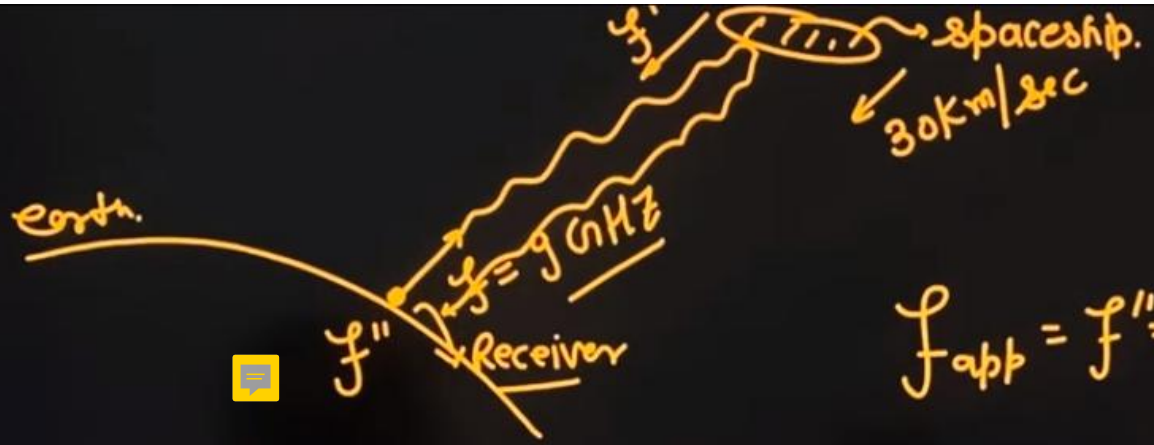
$$= 2.5A^\circ = \frac{v}{c}\lambda$$

$$v = \frac{2.5A^\circ \times c}{\lambda} = \frac{2.5A^\circ \times 3\times10^{8}\ m/s}{5000A^\circ}$$

$$= 1.5\times10^{5}\ m/s = 150\ km/sec$$

competishun

Ex



Find frequency received by receiver.

$$f_{app} = f'' = \frac{c^2}{(c-v)^2} f_0$$

$$= \frac{1}{\left(1-\frac{v}{c}\right)^2} f_0$$

① $$f' = \frac{c}{c-v} f_0$$

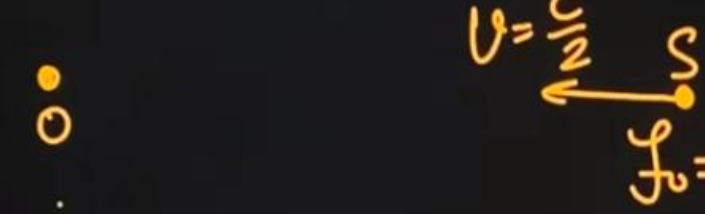
② $$f'' = \frac{c}{c-v} f' = \left(\frac{c}{c-v} \times \frac{c}{c-v}\right) f_0$$

$$f'' = \left(1-\frac{v}{c}\right)^{-2} f_0 = \left(1+\frac{2v}{c}\right) f_0 = \left(1+\frac{2\times 30\times 10^3}{3\times 10^8}\right) \times 9\times 10^9 \text{ Hz}$$

competishun

Ex JEE

c - speed of light.



Source is moving toward observer with speed = $\frac{c}{2}$.

Find frequency observed by observer.

$$f_{app} = \frac{c}{c-v} f_0$$

$\otimes$ $\frac{c}{c/2} f$

f_0 = 20 GHz ✗

$v << c$

$$f_{app} = \sqrt{\frac{c+v}{c-v}}\, f_0 = \sqrt{\frac{c+c/2}{c-c/2}}\, f_0 = \sqrt{3} \times 10\ \text{GHz}$$

$$= 17.32\ \text{GHz}.$$

competishun